AfriLens Report AI

Project Documentation and Technical Report

Prepared By

Team Nessus

Project Title

AfriLens Report AI: AI-Powered Vulnerability Reporting with African Context Intelligence

Date

October 2026

Executive Summary

AfriLens Report AI is an intelligent cybersecurity reporting platform designed to simplify and automate the vulnerability reporting process. The platform allows users to upload vulnerability assessment reports generated by tools such as Nessus and automatically transforms the findings into professional, structured, and actionable reports. The system extracts vulnerabilities, categorizes findings by severity, calculates risk levels, generates recommendations, and exports reports in professional formats.

The unique feature of AfriLens Report AI is its African Context Intelligence capability, which translates technical cybersecurity findings into relatable everyday African experiences and scenarios. This enables students, small businesses, managers, and non-technical stakeholders to better understand cyber risks and the importance of remediation actions. The project addresses the challenge of interpreting complex vulnerability reports and bridges the communication gap between technical and non-technical audiences.

1. Introduction

Cybersecurity professionals rely on vulnerability assessment tools to identify security weaknesses within systems, networks, and applications. While these tools generate valuable information, the resulting reports are often highly technical and difficult for non-security stakeholders to understand. Security analysts also spend significant time manually reviewing findings, prioritizing risks, and preparing reports for decision-makers.

AfriLens Report AI was developed to streamline this process by automatically converting vulnerability scan results into professional reports with clear explanations, risk ratings, remediation guidance, and context-driven interpretations. The platform aims to improve cybersecurity awareness and make vulnerability reporting more accessible and understandable.

2. Problem Statement

Organizations frequently perform vulnerability assessments using tools such as Nessus, but the output generated by these tools is often:

- Complex and highly technical.
- Time-consuming to analyse.
- Difficult for management and non-technical stakeholders to interpret.
- Challenging for students and beginner analysts to prioritize correctly.

As a result, vulnerabilities may not be communicated effectively, remediation efforts may be delayed, and decision-makers may not fully understand the associated risks.

3. Project Objectives

The objectives of AfriLens Report AI are:

- Automate vulnerability reporting.
- Reduce the time required to generate assessment reports.
- Categorize vulnerabilities automatically by severity.
- Generate executive summaries for management.
- Provide remediation recommendations.
- Calculate overall risk scores.
- Translate technical findings into simplified explanations.
- Introduce African-context analogies to improve understanding.
- Export reports in professional formats.

4. Project Scope

The project covers:

- Vulnerability report upload.
- Vulnerability data analysis.
- Risk scoring.
- Severity categorization.
- Automated recommendation generation.

- Executive summary generation.
- African context explanations.
- PDF and DOCX report exports.

The current scope does not include:

- Active vulnerability scanning.
- Penetration testing.
- Automated remediation.
- Threat intelligence integration.
- Real-time security monitoring.

5. Methodology

The project follows the vulnerability management lifecycle:

Step 1: Upload Vulnerability Report

Users upload:

- Nessus files (.nessus)
- CSV files (.csv)
- Excel files (.xlsx)

Step 2: Data Extraction

The system extracts:

- Vulnerability name
- Severity
- CVSS score
- Description
- Host information
- Recommendations

Step 3: Vulnerability Classification

Findings are grouped as:

- Critical

- High
- Medium
- Low
- Informational

Step 4: Risk Scoring

The application calculates risk using predefined severity weights.

Step 5: Report Generation

The system generates:

- Executive Summary
- Findings Summary
- Detailed Findings
- Recommendations
- Conclusion

Step 6: African Context Intelligence

Technical vulnerabilities are translated into relatable real-world African scenarios to improve understanding.

6. System Architecture

Frontend

Provides the user interface for:

- File upload
- Dashboard visualization
- Report viewing
- Report exporting

Technologies:

- HTML
- CSS

- Bootstrap
- JavaScript
- React

Backend

Handles:

- File processing
- Data analysis
- Report generation
- Authentication

Technologies:

- Python
- Flask

Database

Stores:

- Users
- Uploaded reports
- Vulnerability findings
- Generated reports

Technology:

- PostgreSQL

7. Key Features

Vulnerability Upload

Users can upload assessment reports for analysis.

Security Dashboard

Displays:

- Critical findings
- High findings
- Medium findings
- Low findings
- Informational findings

Executive Summary Generator

Automatically creates management-friendly report summaries.

Recommendation Engine

Provides remediation guidance based on findings.

Risk Scoring System

Calculates overall security posture and risk level.

African Context Intelligence

Converts technical cybersecurity concepts into familiar everyday African experiences.

Report Export

Supports:

- PDF Export
- DOCX Export

8. African Context Intelligence

One of the distinguishing features of AfriLens Report AI is the African Context Intelligence module.

This feature helps explain technical vulnerabilities using familiar real-life African situations.

Example 1

Finding: Weak Password Policy

African Explanation:

Using weak passwords is like using the same key for your shop, home, and warehouse. If someone gets a copy of the key, all locations become accessible.

Example 2

Finding: Unsupported Operating System

African Explanation:

Running an unsupported operating system is like operating a matatu for years without maintenance. Eventually, breakdowns become unavoidable and dangerous.

Example 3

Finding: Information Disclosure

African Explanation:

This is similar to leaving your business records displayed openly where anyone passing by can view them.

This approach makes cybersecurity easier to understand for:

- Students
- SMEs
- Managers
- Community organizations
- Non-technical users

9. Results

The project successfully achieved the following:

- Automated vulnerability report processing.

- Dynamic vulnerability categorization.
- Risk score calculation.
- Executive summary generation.
- Recommendation generation.
- Professional report creation.
- African-context explanation generation.

The platform significantly reduces the effort required to transform raw vulnerability findings into actionable reports.

10. Testing Strategy

Functional Testing

Tested:

- File Upload
- Report Analysis
- Dashboard Generation
- Report Export

Result: All core functions performed successfully.

Security Testing

The application was assessed using:

Nessus

Used to identify security weaknesses.

Burp Suite

Used to test:

- Input validation
- Authentication handling
- Upload capabilities

Nmap

Used to identify:

- Services
- Open ports

Wireshark

Used to observe:

- Upload traffic
- Authentication requests
- Downloads

Result: The application functioned as intended and handled uploaded content securely.

11. Challenges Encountered

Several challenges were experienced during development:

- Interpreting different Nessus export formats.
- Structuring report templates consistently.
- Designing effective risk scoring mechanisms.
- Translating technical cybersecurity concepts into meaningful African analogies.
- Ensuring generated reports remain accurate and relevant.

12. Project Impact

AfriLens Report AI offers value to several groups:

Students

Helps beginners understand vulnerability findings.

Cybersecurity Analysts

Reduces manual reporting effort.

Small Businesses

Provides understandable security assessments.

Management Teams

Simplifies cybersecurity decision-making.

Educators

Improves cybersecurity awareness and learning.

13. Limitations

Current limitations include:

- Dependency on uploaded reports.
- Support limited to selected report formats.
- African analogies may not cover every security scenario.
- No live vulnerability scanning capability.
- No threat intelligence integration.

14. Future Improvements

Future versions may include:

- AI-powered remediation prioritization.
- Threat intelligence integration.
- CVE lookups.
- Compliance frameworks integration.
- Multi-language support.
- Cloud deployment.
- Real-time dashboard monitoring.
- Automated trend analysis.
- Multi-user enterprise support.

15. Lessons Learned

The project provided valuable lessons in:

- Vulnerability management.
- Cybersecurity reporting.
- Risk assessment.
- Secure software development.
- User-centered design.
- Effective communication of cyber risks.
- Team collaboration.
- Practical application of cybersecurity principles.

16. Conclusion

AfriLens Report AI successfully addresses the challenge of transforming complex vulnerability assessment results into understandable, actionable, and professional cybersecurity reports. By combining automated reporting, risk analysis, executive summaries, and remediation guidance, the platform helps improve communication between security teams and decision-makers.

The inclusion of African Context Intelligence makes the project unique by translating technical cybersecurity concepts into relatable real-world experiences. This approach improves cybersecurity awareness and accessibility while helping users better understand the significance of security findings. AfriLens Report AI demonstrates how artificial intelligence and automation can simplify cybersecurity workflows and strengthen vulnerability management practices.